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RESEARCH ARTICLE

Advancing AIoMT-Enabled Healthcare System-of-Systems Using Multi-Agent Reinforcement Learning

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ABSTRACT This paper presents a Multi-Agent Reinforcement Learning (MARL) framework designed to optimize coordination within a System-of-Systems (SoS) composed of six heterogeneous healthcare entities: hospitals, clinics, telemedicine platforms, wearable monitoring devices, rural health centers, and virtual triage hubs. These entities are organized into four distinct constituent systems—Directed, Acknowledged, Collaborative, and Virtual—each reflecting a unique coordination model that governs control, autonomy, and communication. Within each constituent system, agents independently learn optimal policies using Q-learning to enhance resource allocation, inter-agent cooperation, and service efficiency. We describe simulation experiments that span 1000 episodes, with evaluation metrics including cumulative rewards, resource utilization, and system-level efficiency. The results show that Acknowledged and Collaborative coordination models achieve faster policy convergence and superior operational outcomes, while the Virtual model provides flexibility with reduced coordination overhead. Our framework demonstrates the potential of MARL to enable adaptive, decentralized decision-making in Artificial Intelligence of Medical Things (AIoMT)-enabled healthcare environments, particularly under dynamic and resource-constrained conditions. Importantly, this approach can help improve care access in remote and underserved regions and enable more dynamic, responsive triage and resource allocation in real-world healthcare delivery.

INDEX TERMS Artificial intelligence of medical things (AIoMT), healthcare systems, multi-agent reinforcement learning (MARL), Q-learning, resource optimization, system-of-systems (SoS).

I. INTRODUCTION

The growing complexity of modern healthcare delivery poses significant integration challenges for systems composed of diverse, autonomous, and distributed entities such as hospitals, clinics, telemedicine units, and connected medical devices. The emergence of the Artificial Intelligence of Medical Things (AIoMT) has introduced new opportunities to enhance coordination, decision-making, and operational efficiency through data-driven intelligence. However, despite advances in digital health technologies, achieving real-time interoperability, scalable resource allocation, and adaptive

collaboration across heterogeneous components remains a persistent challenge.

Healthcare environments increasingly resemble a System-of-Systems (SoS)—a collection of operationally and managerially independent systems that must cooperate to deliver collective value. In this context, traditional rule-based or centralized decision-making approaches are often inadequate to handle the distributed and dynamic nature of healthcare systems. To address these limitations, this paper proposes a Multi-Agent Reinforcement Learning (MARL) framework that enables healthcare subsystems to learn optimal strategies for interaction and coordination over time.

Each agent in the proposed architecture represents an independent healthcare unit (e.g., hospital, telemedicine node, or clinic), capable of making autonomous decisions

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while learning from local observations and feedback. The system is designed to model and compare the performance of four SoS configurations—Directed, Collaborative, Acknowledged, and Virtual—within AIoMT-enabled environments. The goal is to explore how different SoS structures influence system-level performance metrics such as cooperation count, resource utilization, and efficiency under decentralized learning conditions.

The motivation for this study arises from the need to bridge the gap between isolated healthcare systems and develop cohesive, adaptive, and intelligent SoS architectures capable of responding to real-time constraints. MARL, particularly with independent Q-learning agents, offers a lightweight and scalable solution that aligns well with the operational realities of decentralized healthcare environments, including rural and resource-constrained settings.

This paper makes the following key contributions:

- **A modular Multi-Agent Reinforcement Learning (MARL) framework for healthcare Systems-of-Systems (SoS):** We propose a novel SoS architecture that integrates heterogeneous healthcare agents—including hospitals, clinics, telemedicine platforms, and wearable devices—coordinated through MARL to enable adaptive, decentralized decision-making.
- **Comparative analysis of constituent SoS architectures:** The framework systematically models and empirically evaluates four distinct constituent systems—Directed, Acknowledged, Collaborative, and Virtual—enabling a direct comparison of centralized, hybrid, and decentralized coordination strategies within a unified simulation environment.
- **Comprehensive performance assessment:** We define and analyze a suite of system-level metrics—including cooperation count, resource utilization, exploitation/exploration rates, and efficiency—across 1000 simulated episodes, providing new insights into emergent learning and coordination behavior.
- **Demonstration of Q-learning adaptability in distributed healthcare:** Our results demonstrate the adaptability and effectiveness of independent Q-learning agents in resource-constrained and decentralized healthcare environments, highlighting real-world applicability in both rural and under-resourced systems.
- **Reproducible simulation setup and roadmap for real-world impact:** We provide a reproducible simulation framework and discuss the practical implications for AIoMT-based healthcare, including opportunities for deep RL extensions, federated learning, and integration with empirical datasets.

Paper Organization: The remainder of this paper is structured as follows. Section II reviews prior research in AIoMT systems, reinforcement learning in healthcare, and SoS coordination strategies. Section III presents the proposed System-of-Systems (SoS) architecture and its integration with Multi-Agent Reinforcement Learning (MARL). Section IV details the reinforcement learning formulation,

agent definitions, system environment, and evaluation metrics. Section V reports the simulation results across four SoS configurations, with analysis of learning behavior, efficiency, and resource utilization. Section VI discusses practical implications, limitations, and potential extensions. Finally, Section VII concludes the paper and outlines directions for future research.

II. RELATED WORK

A. AIoMT INTEGRATION IN HEALTHCARE SYSTEMS

The emergence of Artificial Intelligence of Medical Things (AIoMT) has transformed healthcare delivery through intelligent sensing, real-time patient monitoring, and personalized treatment recommendations [1], [2]. AIoMT combines artificial intelligence with the Internet of Medical Things (IoMT) to enable remote diagnostics, continuous tracking of physiological parameters, and data-driven clinical decisions [3].

Despite its potential, AIoMT systems remain siloed across hospitals, wearable devices, and telemedicine platforms, often lacking interoperability. Systems-of-Systems (SoS) methodologies offer a promising path toward integrating these heterogeneous components under a unified decision-making architecture [4], [5]. However, practical implementation remains challenging due to communication overhead, data privacy concerns, and the need for scalable decision-making policies.

In summary, the literature demonstrates that while AIoMT has enabled significant advances in sensing and clinical decision support, most implementations remain fragmented and lack true interoperability at scale. Existing work highlights the need for unified, system-level approaches that transcend single-device or institution boundaries—an unmet challenge that motivates our focus on SoS frameworks for coordinated, adaptive healthcare delivery.

B. SYSTEM-LEVEL COORDINATION AND SoS CHALLENGES

Healthcare systems, especially in underserved regions, often suffer from limited infrastructure, lack of standardized protocols, and insufficient training in AIoMT technologies [6], [7]. These challenges highlight the importance of adaptable SoS frameworks capable of functioning in low-resource environments [8]. Recent efforts focus on mathematical and algorithmic integration models to coordinate system interactions in a scalable and privacy-preserving manner [9].

However, SoS-specific frameworks in healthcare remain underexplored. Comparable coordination challenges in constrained domains—such as semiconductor manufacturing—demonstrate how hybrid frameworks combining system-dynamics, optimization, and machine learning can improve decision-making and adaptability [10]. Most research prioritizes device-level intelligence or algorithmic enhancement without fully considering inter-agent autonomy, SoS typologies (e.g., Directed vs. Collaborative), or system-wide cooperation models.

Collectively, prior studies reveal that scalable and adaptive coordination is a recurring challenge, especially in resource-limited settings. However, the literature remains limited in developing or empirically validating system-wide SoS architectures tailored for healthcare. Our work addresses this gap by operationalizing and benchmarking multiple constituent systems within a practical, MARL-enabled SoS environment.

C. MULTI-AGENT REINFORCEMENT LEARNING FOR HEALTHCARE SoS

Reinforcement Learning (RL) is a data-driven machine learning paradigm in which autonomous agents learn to make sequential decisions by interacting with a dynamic environment, with the goal of maximizing long-term cumulative rewards. At each time step, an agent observes its current state, selects an action, and receives a feedback signal (reward), which informs future decisions. Over time, agents refine policies that map states to optimal actions, enabling them to improve their performance through experience.

Multi-Agent Reinforcement Learning (MARL) extends this concept to environments with multiple interacting agents, each learning independently or jointly while influencing one another. MARL offers a dynamic and decentralized approach to healthcare decision-making, enabling agents to adapt based on local observations and shared environments [11]. MARL has shown promising results in disease classification, imbalance mitigation, and personalized treatment plans. However, its application to healthcare SoS remains limited in scope.

Existing work often assumes static coordination settings and does not account for architectural differences across Directed, Collaborative, or Virtual SoS models. Prior studies on circular economy marketplaces reveal that Q-learning and deep Q-learning (DQL) can stabilize pricing and support agent coordination under real-time constraints, providing a valuable precedent for MARL in health SoS settings [12].

Likewise, MARL-enabled digital twins have been used to optimize yield and defect detection in 3D-IC manufacturing SoS by integrating CNNs, Random Forests, and LSTMs for subsystem coordination [13]. These examples indicate the potential of MARL to scale across heterogeneous, distributed, and partially observable systems, though healthcare-specific SoS applications remain underdeveloped.

Recent years have seen significant progress in applying MARL to real-world healthcare problems. For example, explainable MARL frameworks have been used for real-time scheduling and patient flow optimization, integrating human-in-the-loop decision support to reduce wait times and improve adherence to operator constraints [14]. Multi-agent systems have also addressed single-day operating room scheduling [15], hierarchical hospital resource allocation during pandemics [16], and decentralized staff scheduling using conflict resolution protocols [17]. These studies demonstrate that MARL enables adaptive and scalable approaches for

hospital logistics, resource allocation, and dynamic patient triage. Recent advances also include hierarchical MARL for multi-organ disease management [18], adaptive chronic disease management [19], and DRL-driven ambulance deployment to reduce response times in emergency medical systems [20]. However, challenges remain in scaling these approaches to heterogeneous, privacy-constrained healthcare Systems-of-Systems, and further work is needed to ensure robust, interpretable, and ethically sound deployment [21], [22], [23].

Recent advances confirm the potential of MARL for improving scheduling, logistics, and adaptive treatment across healthcare domains. However, the majority of MARL applications target narrow tasks or assume static agent roles, and few address the comparative impact of different SoS architectures or the unique demands of heterogeneous, privacy-constrained environments. Our study explicitly fills this gap by benchmarking multiple SoS coordination models using a reproducible MARL framework.

D. RESEARCH GAP AND CONTRIBUTION RATIONALE

Although the integration of AIoMT technologies and intelligent decision-making has advanced significantly in healthcare, several key gaps persist in current research. These gaps directly motivate the contributions of this study:

- **Lack of holistic SoS integration:** Existing literature often addresses isolated components of healthcare—such as hospitals, wearable devices, or telemedicine platforms—but lacks a unified Systems-of-Systems (SoS) framework to ensure coordinated and interoperable operations [4], [5], [9]. Without such a framework, healthcare systems remain fragmented, limiting their ability to respond adaptively and share resources effectively.
- **Limited application of Multi-Agent Reinforcement Learning (MARL):** While MARL has demonstrated effectiveness in specific healthcare domains like disease prognosis and monitoring [11], its potential remains underutilized in full-scale SoS contexts. Earlier studies on component failure detection using text analytics and multi-agent coordination in electronics manufacturing provide examples of real-time agent collaboration and learning, but have not been extended to healthcare SoS [24].
- **Challenges in scalability and data coordination:** AIoMT-enabled healthcare systems often struggle with data integration across heterogeneous agents, especially in remote or resource-constrained environments. Issues such as infrastructure limitations, privacy concerns, and inconsistent communication protocols hinder effective system-wide coordination [2], [6], [7], [8].
- **Gap in constituent system evaluation within SoS:** While Systems-of-Systems (SoS) models are well-characterized in theory, few studies provide empirical comparisons of distinct constituent systems—such

as Directed, Acknowledged, Collaborative, or Virtual—within a unified SoS framework in healthcare coordination contexts. This limits understanding of how variations in constituent system architecture impact agent behavior, learning performance, and emergent system-level outcomes [25].

To address these gaps, this paper proposes a modular System-of-Systems (SoS) framework for AIoMT-enabled healthcare coordination, driven by Multi-Agent Reinforcement Learning (MARL). The framework models and compares four canonical constituent systems—Directed, Acknowledged, Collaborative, and Virtual—within a unified SoS environment. This enables a deeper understanding of how structural and policy variations across constituent systems affect reward convergence, cooperation, and system efficiency. This work advances scalable, adaptive approaches to AIoMT system design and provides insights for future smart healthcare infrastructures.

III. FRAMEWORK OVERVIEW

To address the coordination, adaptability, and decision-making limitations prevalent in AIoMT-enabled healthcare ecosystems, we propose an integrated *System-of-Systems* (SoS) architecture underpinned by *Multi-Agent Reinforcement Learning* (MARL). This framework enables distributed, goal-driven collaboration among heterogeneous member systems—such as hospitals, clinics, telemedicine platforms, and wearable devices—through dynamic policy learning. A reinforcement learning approach is needed to empower agents to learn optimal strategies over time, adapt to changing conditions, and improve system-level efficiency and care delivery outcomes.

A. SOS ARCHITECTURE FOR AIOMT-ENABLED HEALTHCARE

To explore coordination and learning in complex, decentralized healthcare settings, we design a simulation environment that models AIoMT-enabled ecosystems as a System-of-Systems (SoS). This environment allows us to isolate and evaluate different coordination mechanisms by organizing six heterogeneous agents—hospitals, clinics, telemedicine platforms, and wearable devices—into structurally distinct groups known as *constituent systems*.

The proposed framework includes four constituent systems—*Directed*, *Acknowledged*, *Collaborative*, and *Virtual*—each reflecting a conceptual coordination model commonly found in healthcare integration efforts. The Directed system operates as a rule-based centralized architecture without learning. As such, it serves as a fixed-policy benchmark for evaluating the performance of decentralized, MARL-enabled coordination strategies. While these categories are adapted from general SoS literature [26], [27], we apply them here to represent varying degrees of centralization, autonomy, and data coupling across agents. These models are described below.

- **Collaborative constituent system:** All agents operate autonomously and engage in peer-to-peer coordination. Local decision-making is emphasized, with agents sharing information, negotiating care pathways, and co-adapting based on system feedback. In our implementation, explicit message-passing or structured inter-agent communication protocols were not employed. Instead, agents operate within a shared environment where their decisions influence global resource dynamics and patient distribution. These indirect interactions—such as patient overflow or service backlogs—create emergent environmental feedback that neighboring agents experience through their local states. This setup facilitates peer-aligned adaptation without centralized control, reflecting realistic decentralized coordination in data-rich healthcare systems. This model is well-suited to environments where decentralized response and cross-institutional collaboration are essential.
- **Directed constituent system:** A centralized authority exercises complete control over all agents. Entities receive and execute directives regarding patient routing, staff deployment, and service scheduling. While structured and hierarchical, this model often suffers in responsiveness and adaptability under non-stationary healthcare demands.
- **Acknowledged constituent system:** A hybrid coordination model where agents act autonomously but are guided by high-level goals or policies issued by a central coordinator (e.g., a national health authority). Local decision-making is preserved, but actions are shaped by shared care protocols, enabling a balance between flexibility and system-wide coherence.
- **Virtual constituent system:** Comprised of loosely coupled digital agents—such as wearables and telemedicine services—that communicate via shared data streams rather than through physical infrastructure. Coordination is asynchronous and emergent, prioritizing data-driven triage, remote diagnostics, and continuous monitoring with minimal overhead.

In contrast to the Collaborative constituent system—which relies on direct, peer-to-peer coordination and explicit negotiation among stateful agents—the Virtual constituent system employs stateless, event-driven agents that operate asynchronously. Each agent, such as a wearable or telemedicine platform, acts independently, reacting to updates in a shared cloud-based data environment. Agents do not maintain persistent internal states about the system or other agents; instead, their decisions are triggered by incoming data streams (e.g., patient telemetry, diagnostic alerts). Coordination emerges from the sequence of asynchronous actions and environmental updates, not from explicit inter-agent communication. This architecture enables highly scalable and resilient operations in settings where minimal infrastructure coupling and flexible, real-time responsiveness are critical.

Later in Section IV-D, we describe the simulation architecture used to instantiate each constituent system and evaluate

performance metrics such as cumulative reward, Q-value convergence, and resource efficiency. Figure 1 visually represents the SoS architecture, with agent roles and coordination logic aligned with the entity mappings provided in Table 1.

In each simulation scenario, the same six agents are organized under one of the four constituent systems—Directed, Acknowledged, Collaborative, or Virtual—to enable a direct, comparative evaluation within the overall SoS framework. This design isolates the impact of different coordination architectures on emergent behavior and learning dynamics. Table 1 provides a detailed role mapping for these agents, and the results—such as cumulative rewards, Q-value convergence, cooperation counts, and resource utilization trends—demonstrate how each constituent system influences agent learning, coordination behavior, and operational efficiency in AIoMT-enabled healthcare environments.

Note on Agent Role Consistency: The agent roles and entities assigned in each constituent system (as detailed in Table 1) reflect the operational logic and coordination requirements unique to each SoS architecture. For instance, Virtual and Collaborative systems emphasize decentralized, data-driven coordination among wearables and telemedicine platforms, while Directed and Acknowledged models prioritize hospital- and clinic-centric control. This role adaptation is intentional and necessary to faithfully model the varying structures and communication patterns inherent to real-world SoS integration. However, we acknowledge that this design limits direct, agent-to-agent comparisons across different SoS types. To address this, all performance results are analyzed and discussed within each SoS configuration, rather than as direct comparisons between individual agent types across systems. This approach ensures that observed differences reflect coordination architecture rather than mere agent assignment.

Agent Configuration and Role Assignment: To facilitate fair comparison across constituent systems, each simulation includes six agents fulfilling broadly consistent functional roles—such as patient monitoring, triage, diagnostics, or hospital admission. While the specific entities (e.g., hospitals, wearables, or telemedicine platforms) vary slightly across configurations to reflect the logic of each coordination model, the core role categories are preserved. This design ensures that observed performance differences are attributable to coordination dynamics rather than disparities in agent purpose. Table 1 provides a detailed mapping of agent roles and entities for each constituent system.

These structural assignments anchor each agent's operational context within the broader SoS. Building on this foundation, we now describe how agents apply reinforcement learning to refine their behaviors over time, adapting their strategies based on local feedback and coordination patterns.

B. MULTI-AGENT REINFORCEMENT LEARNING (MARL) INTEGRATION

To optimize resource distribution and adapt to environmental dynamics, each agent employs value-based reinforcement

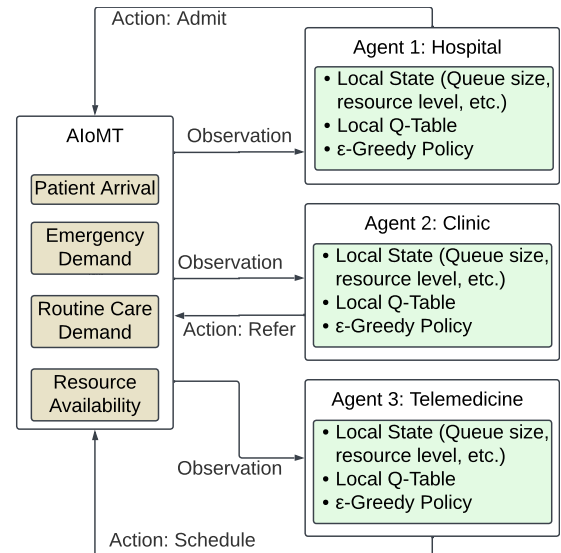


FIGURE 1. System-of-Systems (SoS) architecture for AIoMT-enabled healthcare. Six agents are deployed under four coordination models—Directed, Acknowledged, Collaborative, and Virtual—each shaping interaction patterns, autonomy, and learning behavior.

learning. Specifically, a Q-learning mechanism enables agents to iteratively refine their decision policies through trial-and-error interactions, guided by locally observed states and environmental feedback. This approach allows each agent to autonomously adapt its behavior over time without requiring a complete model of the environment, making it suitable for the stochastic and dynamic conditions typical of healthcare delivery systems.

Each agent plays a distinct role within the healthcare environment. Hospitals are responsible for optimizing bed allocation, personnel deployment, and patient admission. Wearable devices continuously monitor patient vitals, detect physiological anomalies, and communicate status updates to clinical systems. Telemedicine platforms facilitate virtual consultations and remote diagnostics, effectively triaging cases for escalation based on clinical severity and available resources. These functional distinctions ensure that agents collectively span sensing, coordination, and actuation layers of the healthcare System-of-Systems (SoS) architecture.

The high-level SoS architecture used in the AIoMT-enabled healthcare environment is depicted in Figure 1, where agent interactions with the central decision hub are shaped by patient demands, resource availability, and coordination logic. The architecture incorporates four constituent coordination models—Directed, Acknowledged, Collaborative, and Virtual—that define structural constraints on autonomy, communication flow, and decision authority. These models influence emergent system behaviors, including resilience, responsiveness, and policy convergence, as explored in the subsequent performance evaluations.

Each agent seeks to maximize long-term healthcare outcomes by learning policies that balance immediate system demands with strategic performance objectives. This optimization is driven by accumulated experience and

TABLE 1. Role-specific agent configuration across constituent systems.

Constituent System	Agent ID	Entity Type	Primary Role
Directed System	Agent 0	Hospital A	Centralized scheduling and admission
	Agent 1	Hospital B	Diagnostics and resource routing
	Agent 2	Clinic A	Local triage and coordination
	Agent 3	Clinic B	Rural care delivery
	Agent 4	Hospital C	Overflow management
	Agent 5	Hospital D	Resource surplus sharing
Acknowledged System	Agent 0	Hospital A	Shared protocol compliance and patient intake
	Agent 1	Hospital B	Staff and diagnostic optimization
	Agent 2	Clinic A	Emergency services under shared rules
	Agent 3	Clinic B	Peripheral coordination
	Agent 4	Telemedicine Platform	Remote triaging under national policies
	Agent 5	Wearable Node	Vitals reporting under semi-central oversight
Collaborative System	Agent 0	Wearable Device 1	Patient monitoring and alerts
	Agent 1	Wearable Device 2	Continuous vital tracking
	Agent 2	Telemedicine Platform 1	First-line triage
	Agent 3	Telemedicine Platform 2	Cross-agent communication
	Agent 4	Hospital A	Joint planning and admission
	Agent 5	Hospital B	Load balancing and shared diagnostics
Virtual System	Agent 0	Hospital C	AI-guided treatment validation
	Agent 1	Rural Clinic E	Community-level screening
	Agent 2	Clinic B	Minor condition handling
	Agent 3	Telemedicine Hub	Central virtual triage
	Agent 4	Wearable Node 1	Vital collection and stream transmission
	Agent 5	Wearable Node 2	Monitoring and alert escalation

shaped by the degree of agent interdependence, ultimately supporting scalable and intelligent decision-making across the distributed SoS environment.

IV. METHODOLOGY

This section outlines the methodological foundation of the proposed Multi-Agent Reinforcement Learning (MARL) framework, developed to enhance coordination, resource allocation, and patient care across AIoMT-enabled healthcare Systems-of-Systems (SoS). The framework leverages a value-based reinforcement learning approach—specifically, the Q-learning algorithm—across four distinct SoS coordination structures: Directed, Acknowledged, Collaborative, and Virtual. Each architecture represents a unique organizational paradigm in distributed healthcare environments, reflecting varying levels of autonomy, control, and communication among constituent systems.

In this formulation, each agent (e.g., hospital, telemedicine unit, or wearable device) learns an individualized policy to optimize local decisions while contributing to emergent system-level behavior. The learning process unfolds over a sequence of episodic interactions within a simulated environment, where agents adapt their strategies in response to stochastic state transitions and feedback signals encoded as rewards. This decentralized structure aligns with principles of distributed decision-making and enables the emergence

of adaptive behavior, particularly under conditions of uncertainty and partial observability typical in real-world healthcare systems. The reinforcement learning model captures dynamic interdependencies across agents, allowing the system to evolve policy strategies that balance local efficiency with global coordination.

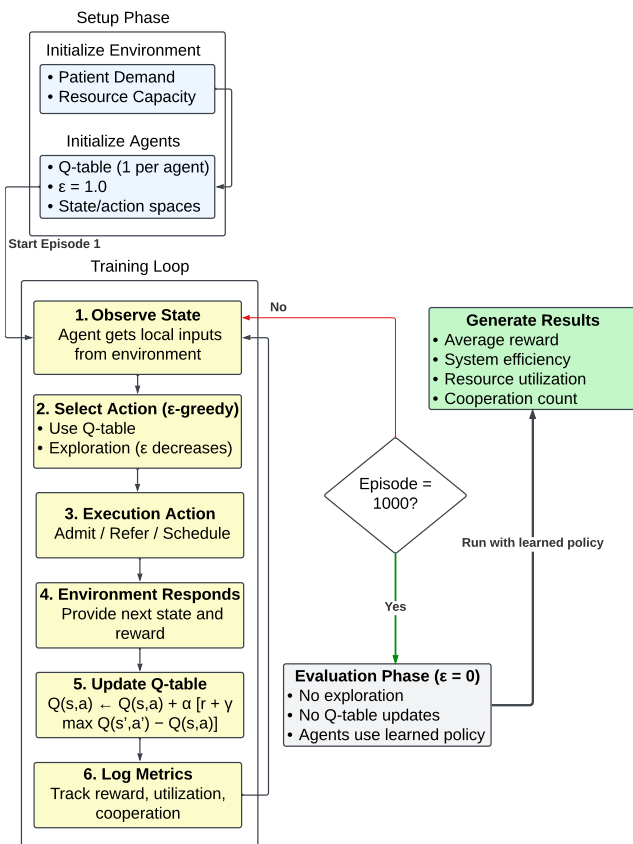
The training pipeline, illustrated in Figure 2, begins with exploration-driven interactions governed by an ϵ -greedy policy. Over the course of 1000 episodes, agents incrementally update their Q-values, refining their policies based on observed outcomes and peer behaviors. Upon convergence, the system enters an evaluation phase ($\epsilon = 0$), during which exploitation dominates and the effectiveness of learned behaviors is assessed across multiple performance dimensions. These include cumulative rewards, cooperation levels, system efficiency, and resource utilization metrics—each of which is subsequently analyzed to compare emergent behavior across the constituent system configurations modeled within the overall SoS framework.

A. REINFORCEMENT LEARNING FRAMEWORK AND Q-LEARNING ALGORITHM

The formulation below assumes basic familiarity with reinforcement learning and Markov decision processes (MDPs); for foundational treatments, see [28], [29].

TABLE 2. Constituent system breakdown in the proposed MARL-Based healthcare framework.

Constituent System	Coordination Model	Control Flow	Example Behavior in AIoMT-Enabled Healthcare System
Directed System	Centralized	A single central authority directs all decision-making, resource distribution, and agent behavior.	A lead hospital acts as the orchestrator, assigning patient loads, staff resources, and intervention schedules to other entities (clinics, telemedicine platforms), with agents executing rather than planning actions.
Acknowledged System	Semi-centralized	Agents operate independently but follow shared policies and periodically synchronize with a coordinating entity.	Hospitals, clinics, and remote care units independently optimize patient flow and staffing while complying with national care protocols and data-sharing policies for coordinated goals.
Collaborative System	Decentralized	All agents have equal autonomy and achieve coordination through negotiation and peer-to-peer communication.	Regional clinics and hospitals share diagnosis data and negotiate patient transfers or load balancing, with no single point of authority, leading to emergent coordination in crisis response.
Virtual System	Loosely coupled, asynchronous	Coordination arises from real-time data streams; agents are digital and may never be physically integrated.	Wearables continuously stream health metrics to AI-enabled cloud systems. Telemedicine agents asynchronously analyze the data, trigger alerts, and suggest interventions without centralized scheduling.

**FIGURE 2.** Training and evaluation process for the proposed MARL framework. Agents interact with the environment using an ϵ -greedy policy over 1000 episodes. After training, an evaluation phase (with $\epsilon = 0$) is used to assess learned agent behavior and system performance, including reward, efficiency, resource utilization, and cooperation count.

To enable adaptive decision-making in dynamic healthcare environments, each agent within the proposed Systems-of-Systems (SoS) framework—including those in Directed,

Acknowledged, Collaborative, and Virtual configurations—employs a value-based reinforcement learning technique—Q-learning. This algorithm allows agents to iteratively estimate the expected utility of actions taken in specific system states, progressively refining their policies based on interaction with the environment. In doing so, agents strive to maximize long-term cumulative rewards that reflect desirable healthcare outcomes such as efficient resource usage and improved patient satisfaction.

Tabular Q-learning was selected due to its interpretability and computational tractability within the bounded simulation environment used in this study.

The core of the Q-learning algorithm is the update of the action-value function $Q(s, a)$, which estimates the expected return for selecting action a in state s , and then following the agent's current policy thereafter. The update rule is defined as [30]:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R(s_t, a_t) + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \quad (1)$$

where $\alpha \in (0, 1)$ is the learning rate that determines the weight given to new experiences, $R(s_t, a_t)$ is the reward received after executing action a_t in state s_t , and $\gamma \in [0, 1]$ is the discount factor controlling the importance of future rewards relative to immediate gains. The term $\max_{a'} Q(s_{t+1}, a')$ represents the maximum Q-value associated with the next state s_{t+1} , guiding the agent toward optimal future behavior.

As agents explore the environment and observe the consequences of their actions, they must strike a balance between exploring novel strategies and exploiting known successful policies. This trade-off is governed by an ϵ -greedy exploration mechanism, where the action a_t at time t is

selected according to [28]:

$$a_t = \begin{cases} \text{random action} & \text{with probability } \epsilon, \\ \arg \max_{a'} Q(s_t, a') & \text{with probability } 1 - \epsilon. \end{cases} \quad (2)$$

Initially, the exploration rate ϵ is set to a high value to encourage diverse action sampling. Over time, ϵ is decayed gradually—typically using a decay schedule of the form $\epsilon_t = \epsilon_0 \cdot e^{-\lambda t}$ —which reduces random action selection and promotes reliance on learned policies.

This reinforcement learning framework enables each agent to autonomously discover effective behavior in the face of non-stationary dynamics and partial observability typical of healthcare environments. By learning from rewards that encode both local and system-wide objectives, agents adapt to diverse operational contexts and contribute to the emergence of coordinated, high-efficiency behavior within the overall SoS.

Algorithm 1 Q-Learning Process for Healthcare SoS Agents

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1: for each episode  $e = 1$  to  $N$  do
2:   for each agent  $i$  do
3:     Initialize state  $s_i$ 
4:     repeat
5:       Choose action  $a_i$  using  $\epsilon$ -greedy policy
6:       Execute action  $a_i$ , receive reward  $r_i$  and observe
         next state  $s'_i$ 
7:       Update Q-table:  $Q(s_i, a_i) \leftarrow Q(s_i, a_i) + \alpha[r_i +$ 
          $\gamma \max_{a'} Q(s'_i, a') - Q(s_i, a_i)]$ 
8:        $s_i \leftarrow s'_i$ 
9:     until terminal state is reached
10:   end for
11: end for

```

B. SYSTEM ENVIRONMENT AND AGENT DEFINITIONS

Each constituent system modeled within the proposed System-of-Systems (SoS) framework is formulated as a Markov Decision Process (MDP), which provides a formal structure for capturing the interaction between agents and their dynamic healthcare environment [29]. The MDP for each constituent system is defined by the tuple (S, A, P, R) , where S represents the state space, A the action space, P the transition probability function, and R the reward function. These components collectively enable agents to learn optimal policies through experience-based adaptation.

Each constituent system—Directed, Acknowledged, Collaborative, and Virtual—features a distinct composition of six agents to reflect realistic architectural differences in coordination strategies. For example, the Collaborative and Virtual systems include wearable and telemedicine agents more prominently, whereas the Directed and Acknowledged systems emphasize hospitals and clinics. Table 1 outlines the specific agent assignments associated with each constituent system configuration.

The state space S captures the evolving system context at each time step t , encapsulating variables such as resource

availability R_t , patient conditions P_t , and diagnostic statuses D_t . Each agent i has access to a localized subset of the global state, denoted as $s_t^i \subset S$, which reflects its individual operational context within the broader SoS environment.

Note on State Space Discretization: To ensure uniform agent architecture and computational tractability, each agent's local state space is discretized into a synthetic 10-state Markov Decision Process (MDP). These states do not directly correspond to interpretable healthcare categories such as triage level or queue length. Instead, they abstractly represent clusters of environmental and patient-related variables—such as resource availability, diagnostic uncertainty, and patient condition—which vary across episodes and agent roles. The states are initialized randomly at the beginning of each episode and evolve stochastically based on agent interactions and dataset-driven transitions. This modeling abstraction supports efficient training while preserving behavioral diversity across agents and SoS configurations.

The action space A comprises the set of possible decisions available to each agent. At every time step t , agent i selects an action $a_t^i \in A$, which may correspond to operations such as admitting or redirecting patients, allocating medical staff, scheduling diagnostic services, or adjusting remote monitoring protocols. The collective actions influence both local outcomes and system-level dynamics.

Transitions between states are governed by the stochastic transition function $P(s_{t+1}|s_t, a_t)$, which captures the probability of transitioning from state s_t to state s_{t+1} given action a_t . These transitions reflect the inherent uncertainty and variability in healthcare operations, such as fluctuations in patient arrival rates or equipment availability. In our implementation, these dynamics are simulated using randomized selections from a synthetically generated dataset, allowing controlled variation while maintaining realism.

The reward function $R(s_t, a_t)$ quantifies the immediate performance feedback associated with a given action a_t in state s_t . This feedback is aligned with critical healthcare objectives, including reducing patient wait times, optimizing resource utilization, and improving overall system responsiveness [31], [32].

Each agent seeks to maximize its expected cumulative reward over a finite time horizon T , with future outcomes discounted by a factor $\gamma \in [0, 1]$. Formally, the objective for agent i is to maximize the value function [28]:

$$J(\pi^i) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t^i) \right], \quad (3)$$

where π^i is the policy followed by agent i , and $\mathbb{E}$ denotes the expectation over all possible trajectories governed by the stochastic transition dynamics.

The discount factor γ is calibrated to prioritize short-term healthcare decisions while still valuing future benefits. A value close to 1 encourages long-term planning, whereas a smaller value biases decision-making toward immediate outcomes. This formulation supports adaptive behavior in

both acute and chronic care scenarios, enabling agents to learn policies that balance present needs against evolving system states.

Policy Framing: At each episode, agents are presented with an observed state constructed from the underlying synthetic dataset—specifically, attributes such as patient demographics, diagnoses, device types, and real-time facility resource availability. The agent’s policy is formally defined as a mapping from these observed states to an action (admit, reroute, allocate resource), and is iteratively improved via Q-learning based on reward feedback derived from resource consumption and patient outcome data. In this manner, the state-action-reward tuple underlying the policy is directly instantiated from the dataset described in Section V. This ensures that the agent’s learned policy remains grounded in realistic, data-driven scenarios throughout the training process.

C. PERFORMANCE METRICS AND EVALUATION

The evaluation of the proposed framework is based on a comprehensive set of individual and system-level metrics designed to capture the performance, adaptability, and coordination quality of agents operating under different constituent systems within the overarching System-of-Systems (SoS) architecture. These metrics serve as the foundation for analyzing emergent behavior across the simulation horizon and support comparative assessments of the Directed, Acknowledged, Collaborative, and Virtual constituent systems [33].

To capture per-episode reward dynamics across agents and episodes, we define three scalar summary statistics: the minimum, maximum, and average total reward for each episode t , denoted respectively as $R_{\min}^t$, $R_{\max}^t$, and $\bar{R}^t$. These are computed as:

$$R_{\min}^t = \min_i \left(\sum_{\tau=1}^T r_{\tau}^i \right), \quad (4)$$

$$R_{\max}^t = \max_i \left(\sum_{\tau=1}^T r_{\tau}^i \right), \quad (5)$$

$$\bar{R}^t = \frac{1}{|A|} \sum_i \sum_{\tau=1}^T r_{\tau}^i, \quad (6)$$

where r_{τ}^i is the reward received by agent i at time step τ during episode t , and $|A| = 6$ is the total number of agents. These three values are visualized in the Results section as episodic reward trend curves to evaluate the consistency and robustness of learning across agents.

At the individual level, each agent accumulates rewards across episodes based on the effectiveness of its decisions. The cumulative reward, denoted R_{total}^i , reflects the agent’s ability to align its actions with system objectives, particularly under conditions of constrained resource availability. To understand how agents balance the trade-off between exploitation and exploration, two behavioral indicators

are monitored: the count of exploitative actions (where the agent selects the action that maximizes the Q-value) and the count of exploratory actions (where the agent selects a random action). These behavioral patterns provide insight into the stability and convergence of learned policies.

Cooperation counts are also recorded, capturing the frequency of interactions between agents that contribute to shared goals—such as transferring patients or sharing diagnostic feedback. These are especially informative in Collaborative and Acknowledged SoS configurations, where inter-agent coordination is structurally encouraged. Additionally, resource utilization U_t^i is measured for each agent at each time step t , quantifying how effectively available resources (e.g., hospital beds, staff, and diagnostic devices) are allocated over time.

At the system-wide level, aggregated metrics provide a macro-level perspective of overall system dynamics. One key metric is the system efficiency η_t , defined as the ratio of successful care delivery events (e.g., treated patients or resolved diagnostic queries) to the total resources consumed [32]:

$$\eta_t = \frac{O_t^{\text{successful}}}{R_t^{\text{total}}} \quad (7)$$

where $O_t^{\text{successful}}$ denotes the number of successful outcomes at time t , and R_t^{total} represents the aggregate resource consumption by all agents during the same interval.

The overall cooperation level is derived from the mean cooperation counts across all agents, providing a scalar measure of emergent collaborative behavior within the SoS. Similarly, exploration rates are computed as the average frequency of exploratory actions across all agents and episodes, revealing the adaptability of each system configuration in the face of dynamic uncertainties.

To enable a clearer interpretation of episodic trends, especially under high-frequency variance, all time-series metrics are post-processed using a moving average filter. The smoothed value $\hat{X}_t$ for a given metric X at episode t is calculated as [34]:

$$\hat{X}_t = \frac{1}{w} \sum_{i=t-w}^t X_i \quad (8)$$

where w is the smoothing window size and X_i is the raw metric value at episode i . This smoothing process preserves long-term dynamics while suppressing short-term noise and aligns with the visualizations presented in the Results section.

Collectively, these metrics establish a robust foundation for evaluating how different coordination structures influence agent learning, resource efficiency, and collaborative behavior within AIoMT-enabled healthcare Systems-of-Systems. These evaluation criteria are applied to the simulation outputs discussed in Section V.

D. EVALUATION PROCESS

We evaluate the proposed MARL framework is conducted through extensive simulations across the four defined Systems-of-Systems (SoS) configurations: Directed, Acknowledged, Collaborative, and Virtual. Each configuration is simulated independently to preserve the structural distinctions in coordination and control, enabling a comparative analysis of emergent system behaviors. The training process spans a total of $N = 1000$ episodes per simulation, during which agents iteratively learn optimal strategies through reinforcement-based interactions with a dynamic healthcare environment.

At the beginning of each simulation, all agents are initialized with random Q-values, and the system state s_0 is sampled from a synthetic distribution representing real-world healthcare inputs, including patient acuity levels, diagnostic uncertainty, and heterogeneous resource availability. This initialization ensures variability and generalizability across different starting conditions.

During each episode $t \in [0, N]$, agents observe their current state s_t , select actions a_t using the ϵ -greedy policy defined in (2), receive corresponding rewards $r_t = R(s_t, a_t)$, and update their Q-values using the learning rule in (1). The cumulative reward over time is computed for each agent to assess policy convergence [28]:

$$R_{\text{cumulative}}^i = \sum_{t=0}^N r_t^i \quad (9)$$

where $R_{\text{cumulative}}^i$ represents the cumulative reward accrued by agent i across the simulation horizon. In parallel, cooperation counts C_t^i , exploitation rates ξ_t^i , and exploration frequencies ϵ_t^i are tracked to characterize agent learning dynamics.

After each episode, both individual-level and system-level performance metrics are recorded. These include smoothed Q-values $\hat{Q}_t$, normalized system efficiency η_t , and aggregated resource utilization U_t , with smoothing applied using a moving average filter:

$$\hat{X}_t = \frac{1}{w} \sum_{k=t-w}^t X_k \quad (10)$$

where X_k denotes the metric at episode k , and w is the window size used to suppress short-term fluctuations while preserving long-term trends. These filtered metrics are used in the generation of convergence plots and behavioral trend analyses as shown in Section V.

For final comparative evaluation, each SoS configuration is also summarized using episode-averaged values of key performance indicators: total cumulative reward R_{total}^i , average Q-value $\bar{Q}^i$, mean cooperation count $\bar{C}^i$, average resource utilization $\bar{U}^i$, and exploration-exploitation rates. These are computed over the final evaluation window $E \subseteq [t_{\text{eval-start}}, t_{\text{end}}]$, with:

$$\bar{Q}^i = \frac{1}{|E|} \sum_{t \in E} Q_t^i, \quad \bar{U}^i = \frac{1}{|E|} \sum_{t \in E} U_t^i, \quad (11)$$

$$\bar{C}^i = \frac{1}{|E|} \sum_{t \in E} C_t^i, \quad \text{Exploit}_i = \frac{\sum_{t \in E} \mathbb{I}_{\text{exploit}}^i(t)}{|E|}, \quad (12)$$

where $\mathbb{I}_{\text{exploit}}^i(t)$ is an indicator function equal to 1 if agent i selected an exploitative action at time t . These final metrics are used to construct comparative charts, as shown in the Results section.

The final phase involves a comparative synthesis across Systems-of-Systems (SoS) configurations, focusing on how coordination structures shape the emergent behavior of trained agents. Key aspects of this analysis include cooperation dynamics, the balance between exploitation and exploration strategies, and the overall efficiency of system-level decision-making. This synthesis provides insights into how varying degrees of autonomy, centralization, and communication influence adaptive learning and performance in distributed healthcare environments.

To assess scalability and robustness, additional simulation runs are conducted by varying the number of agents $|A|$ and perturbing environment complexity, such as through stochastic surges in patient demand or sudden reductions in resource capacity. These scenarios allow for stress testing the adaptability of each SoS model under conditions of operational uncertainty.

V. RESULTS

A. DATASET OVERVIEW

The dataset underpinning this study simulates a multi-agent healthcare system by integrating synthetic patient records, facility-level resources, and diagnostic outcomes. This enables robust modeling of decision-making and resource allocation within the proposed Multi-Agent Reinforcement Learning (MARL) framework.

The patient dataset comprises 500 synthetic records, with an average age of 53.9 years and a near-equal gender distribution (51% male, 49% female). Diagnoses span common chronic and acute conditions such as asthma, hypertension, and cardiovascular disease. Care requirements range from routine to emergency, supporting diverse operational scenarios for agent training.

Resource data represent six healthcare facilities—including hospitals and telemedicine units—each initialized with an average of 103 beds and 85 personnel. While not explicitly modeled, real-world constraints such as diagnostic throughput and infrastructure capacity are implicitly captured through episodic state transitions and reward feedback in the environment.

Additionally, 500 diagnostic outcome records are synthesized, reflecting an average diagnostic accuracy of 89.9% when assisted by AI tools. These outcomes model the integration of intelligent diagnostic support into care workflows and inform reward structures within the MARL training loop.

All data are synthetically generated to emulate realistic healthcare interactions while ensuring compliance with ethical standards and privacy constraints. This synthetic dataset enables reproducible experimentation and scalable

simulation of healthcare Systems-of-Systems under MARL control.

Policy Framing and Dataset Utilization: The synthetic dataset forms the basis for the entire agent training process. At the start of each simulation episode, agents are presented with patient records and facility-level resource states sampled from these data distributions. The agent's observed state encapsulates a composite of patient attributes (age, diagnosis, device type), resource availability (beds, staff), and current facility status. These observed states drive the decision process, with each agent independently selecting actions—such as admitting or rerouting a patient—using the Q-learning policy described in Section IV.

The reward function is informed by diagnostic outcomes and resource consumption in the data, providing feedback for every action taken. This closed loop between data-driven state representation, policy selection, and reward-based adaptation is central to training agents to develop optimal decision policies over the simulation horizon. Thus, the dataset is not only used for environment initialization but is integral to the ongoing reinforcement learning and policy refinement process for each agent.

B. ENVIRONMENT SETUP

The simulation environment is structured to model decision-making processes in a healthcare Systems-of-Systems (SoS) using reinforcement learning, specifically the Q-Learning algorithm. The environment consists of six healthcare agents—hospitals, clinics, and telemedicine platforms—tasked with making binary decisions (admit or reject patients) based on their current states and learned policies.

The simulation spans **1000 episodes**, during which the agents aim to optimize cumulative rewards, representing improved patient outcomes and efficient resource utilization. Each agent operates in a **10-state Markov Decision Process (MDP)**. The states are initialized randomly and updated at each episode to simulate changing conditions such as patient demand, resource availability, and hospital capacity.

Key Q-Learning parameters were defined as follows:

- **Discount factor** ($\gamma = 0.95$): This factor controls how much future rewards influence current decisions, with a high value emphasizing long-term outcomes.
- **Learning rate** ($\alpha = 0.01$): This controls the step size in the Q-value update, ensuring a smooth adjustment of learned policies over time.
- **Exploration rate** ($\epsilon = 0.1$): Using an epsilon-greedy policy, each agent selects a random action with 10% probability (exploration) and chooses the action with the highest Q-value with 90% probability (exploitation), balancing the trade-off between exploring new strategies and exploiting known ones.

The exploration rate ϵ was not held constant during training. Instead, it decayed exponentially from 1.0 to 0.1 across episodes using the schedule $\epsilon_t = \epsilon_0 \cdot e^{-\lambda t}$, where

$\lambda = 0.003$. This allowed agents to explore more broadly during early training while gradually converging to stable policies over time.

Each agent's decision-making is driven by its Q-table, initialized to zero for all state-action pairs. The Q-table is a **10×2 matrix** for each agent, where rows correspond to the 10 possible states and columns represent the 2 actions (admit or reject). Q-values are updated during each episode based on the rule described in (1), which incorporates the immediate reward r , the discount factor γ , and the learning rate α .

To simulate healthcare decision-making under realistic yet controlled conditions, each episode includes a synthetic population of patient profiles. These patients vary by age (ranging from 18 to 90), gender, diagnosis category (e.g., asthma, diabetes), and assigned device type (e.g., wearable BP monitor, pulse oximeter). Patients interact with agent types including hospitals, clinics, telemedicine platforms, and wearables. While not all features are explicitly used in decision-making, the attributes reflect heterogeneity consistent with operational healthcare settings. The synthetic distributions were manually designed, and future work will integrate real-world patient statistics to validate and enhance simulation fidelity.

In the simulation, all resource consumption events—such as admitting a patient, allocating staff, or utilizing a device—are treated with equal weight across agent types. That is, each utilization event increments the relevant resource count by one, regardless of whether the action is performed by a hospital, clinic, or telemedicine agent. This equal-cost abstraction allows for a transparent comparison of coordination strategies without introducing variability due to heterogeneous resource costs. While this approach is effective for isolating learning and cooperation effects, we acknowledge that real-world healthcare environments exhibit more complex and context-specific resource valuation. Future work will extend the framework to include differentiated cost models for various agent and resource types, enabling more granular analysis of fairness and efficiency in system-wide resource allocation.

Rewards are generated randomly for each agent in the range of **[−10, 10]** to simulate the stochastic nature of patient outcomes and resource management.

Reward Sampling Rationale: The interval $[−10, +10]$ was selected to reflect the range of plausible outcomes in operational healthcare scenarios, such as successful interventions, resource strain, or treatment delays. This randomization introduces bounded stochasticity without relying on hard-coded outcome models, thereby enabling more generalizable and architecture-agnostic training. While this design may introduce some variability, it also enables agents to robustly learn policies under uncertainty—a key feature of real-world healthcare settings. To ensure stable convergence despite the randomness, all episodic trends are smoothed and evaluated over a long training horizon of 1000 episodes.

This bounded randomness reflects real-world variability in healthcare outcomes and is smoothed over 1000 episodes to

support convergence, as captured in the reward accumulation and Q-value trends.

At the end of each episode, the cumulative reward for each agent is recorded and used to assess overall performance across the simulation.

This setup allows agents to iteratively improve their decision-making policies by adjusting their Q-values based on immediate and future rewards, with the aim of optimizing resource allocation and patient care over time. The following subsection will present the results of the simulation, including cumulative reward trajectories and Q-value evolution for each healthcare agent. For evaluation purposes, we define a cooperation event as any instance where an agent reroutes a patient to another facility due to capacity limitations or policy constraints. The “cooperation count” metric reflects the cumulative number of such reroute-based interactions, serving as a proxy for agent-level coordination behavior across the simulation horizon.

Although the simulation framework can be extended to accommodate more agents, the current study was limited to six agents due to computation and runtime constraints. Evaluating scalability with larger agent counts (e.g., 12 or 18 agents) remains an important direction for future work.

C. SYSTEMS-OF-SYSTEMS (SoS) INTEGRATION AND AGENT PERFORMANCE

This section presents a comparative evaluation of system integration, agent learning behavior, coordination effectiveness, and system-level efficiency across the four constituent SoS configurations: Directed, Acknowledged, Collaborative, and Virtual. All simulations were executed deterministically using fixed random seeds and synthetic inputs. Episode-level performance metrics were smoothed using a moving average filter to suppress short-term noise and highlight long-term learning dynamics. Specifically, we applied a simple moving average with a window size of 50 episodes; each plotted point thus represents the average of the current and 49 preceding episodes. This approach reduces episodic volatility and clarifies convergence behavior over time.

As a result, variability is conveyed through aggregate metric trends (e.g., reward, Q-value) rather than explicit statistical error bars. *Standard statistical error bars are not included because episode-level metrics are temporally correlated and further smoothed via the moving average; thus, conventional error bars would not reflect true variance and could be misleading.*

Each sub-analysis provides insight into how coordination models shape agent adaptation and overall system performance in the simulated healthcare environment.

Note on Smoothing and Variability: All episodic performance curves shown in this section (e.g., rewards, Q-values, cooperation) were derived from a single deterministic simulation run per SoS configuration using fixed random seeds and synthetic inputs. To emphasize long-term convergence trends over short-term fluctuations, each curve was smoothed using a simple moving average filter with a window size

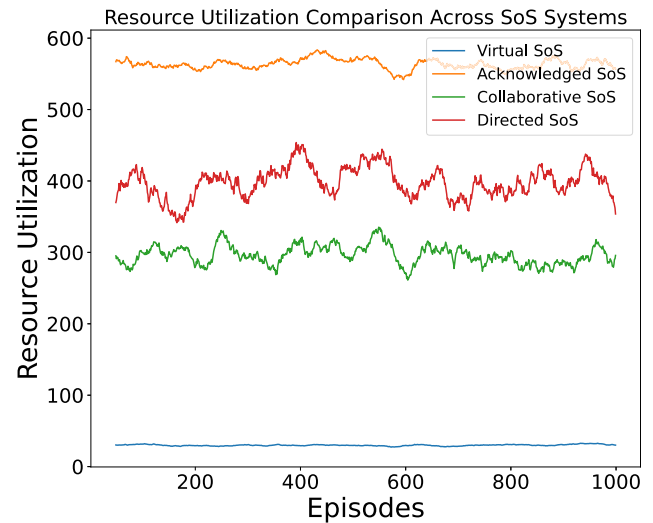


FIGURE 3. Resource utilization trends across four SoS systems over 1000 episodes. The Acknowledged SoS maintains the highest average utilization, followed by Directed and Collaborative systems, while Virtual SoS exhibits minimal physical resource use due to its reliance on telemedicine infrastructure. These patterns reflect each system's efficiency in deploying healthcare resources under varying coordination structures. *Note: The curves are smoothed using a moving average filter applied to a single deterministic run. Variability is visualized through trend dynamics rather than error bars.*

of 50 episodes. As a result, these plots do not include statistical error bars or confidence intervals. While this design supports reproducibility and visual clarity, future work will incorporate multi-run variance analysis to provide deeper insight into performance robustness across stochastic trials.

1) SYSTEMS INTEGRATION

In this section, we analyze the comparative efficiency of resource utilization across four Systems-of-Systems (SoS) architectures: Directed, Acknowledged, Collaborative, and Virtual. Figure 3 illustrates resource consumption trends (beds, staff, and devices) over 1000 episodes for each system. The Acknowledged SoS exhibits the highest resource utilization, averaging approximately 580 units, demonstrating superior coordination and resource optimization across decentralized healthcare entities. In contrast, the Directed SoS shows fluctuating utilization between 380 and 450 units, reflecting its centrally governed resource allocation strategy.

The Collaborative SoS maintains resource utilization between 290 and 340 units, suggesting that peer-to-peer cooperation enhances efficiency, though its scalability remains less than that of the Acknowledged system. The Virtual SoS consistently shows the lowest utilization, below 100 units, due to the reduced demand for physical resources in telemedicine environments.

These trends reveal distinct resource management strategies: Acknowledged SoS maximizes resource allocation through decentralized cooperation, while Virtual SoS minimizes physical resource use, tailored to remote care.

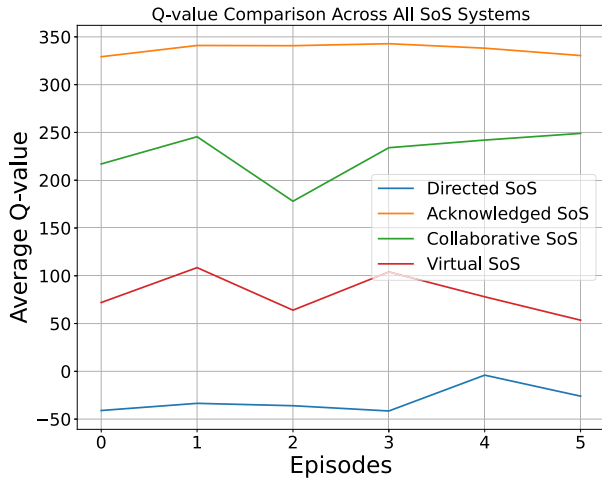


FIGURE 4. Average Q-value comparison across all SoS systems. The Acknowledged SoS system achieves the highest Q-values with minimal variation, indicating effective policy learning. The Collaborative SoS exhibits fluctuating performance with moderate Q-values, while the Virtual SoS maintains stable but lower Q-values. The Directed SoS shows consistently negative Q-values, suggesting limited learning under centralized control. *Note: Q-value trends are smoothed using a moving average filter based on a single deterministic run; statistical error bars are not shown.*

2) AGENT PERFORMANCE AND Q-LEARNING ADAPTATION

This section analyzes the adaptation and learning behavior of agents within the Directed, Acknowledged, Collaborative, and Virtual SoS systems using Q-learning. Key metrics include average Q-values and cumulative rewards, which indicate the agents' performance and decision-making efficiency over time.

Average Q-values across episodes for each SoS system, as shown in Figure 4, reveal distinct learning behaviors shaped by their respective coordination structures. The Acknowledged SoS achieves the highest and most stable Q-values (approximately 330 to 340), reflecting effective policy learning. The Collaborative SoS exhibits greater fluctuation, with values ranging from 180 to 240, indicating variable performance under decentralized control. The Virtual SoS maintains consistent but moderate Q-values (60 to 110), constrained by limited agent interactions. In contrast, the Directed SoS performs poorly, with Q-values ranging from -50 to -10, underscoring persistent learning challenges under centralized authority.

The reward dynamics of the Acknowledged SoS system, depicted in Figure 5, demonstrate consistently strong performance over 1000 episodes. Total rewards typically range between 15 and 25, while the average reward curve stabilizes around 3.5. Maximum rewards remain just below 5 throughout, indicating robust system-wide effectiveness. However, the minimum reward curve shows more pronounced fluctuations—from 0 to 4—suggesting that some agents occasionally face suboptimal conditions despite the system's overall reliability.

In the Collaborative SoS system, reward metrics exhibit notable variability, as shown in Figure 6. Total and maximum

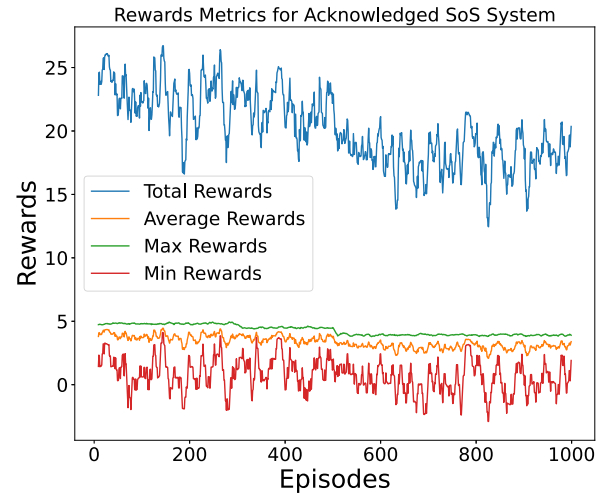


FIGURE 5. Reward metrics for the Acknowledged SoS system. Total, average, and maximum rewards remain consistently high and stable across episodes, indicating effective policy convergence. Fluctuations in minimum rewards suggest intermittent suboptimal behavior by individual agents. *Note: Results are smoothed using a moving average filter over 1000 episodes from a single deterministic run; statistical variability is not shown.*

rewards fluctuate significantly—from 0 to over 40 units—reflecting the dynamic and decentralized nature of agent coordination. Despite these swings, average rewards hover steadily around 5 units, suggesting a stable baseline. Minimum rewards, however, frequently fall below zero, revealing moments of local inefficiency due to transient coordination misalignments.

Reward patterns for the Directed SoS system, illustrated in Figure 7, reveal systemic instability. Total rewards span a wide range from -15 to +15 units, and average rewards oscillate around 0, indicating limited policy learning and poor convergence. While maximum rewards occasionally peak near 10 units, the minimum rewards remain persistently negative, pointing to consistent underperformance rooted in the rigidity of centralized control. To improve trend visibility, the plotted curves in Figure 7 were smoothed using a moving average filter.

As depicted in Figure 8, the Virtual SoS system exhibits moderate but stable reward dynamics. Total rewards range between 0 and 20 units, while average rewards stabilize around 2. Maximum rewards show frequent variation, reaching up to 15 units, whereas minimum rewards remain consistently negative—often dropping below -5. These patterns underscore the inherent limitations of asynchronous, data-driven coordination in delivering consistently high agent-level performance.

Each constituent system configuration utilizes a distinct set of six agents tailored to its specific coordination model (as detailed in Table 1). Accordingly, cumulative reward plots illustrate intra-system performance and are not intended for direct cross-agent comparisons between constituent systems.

In the Acknowledged SoS system, cumulative reward trajectories for all agents show clear and consistent

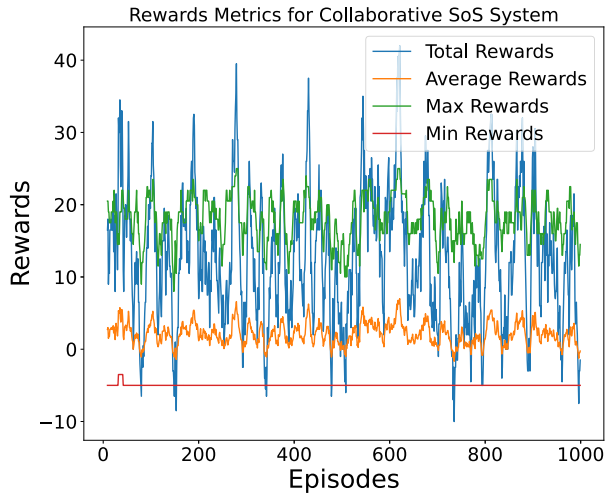


FIGURE 6. Reward metrics for the Collaborative SoS system. High variance in total and maximum rewards reflects the dynamic interactions typical of decentralized decision-making. Occasional sharp drops in minimum rewards highlight local coordination challenges despite stable average performance. *Note: Results are smoothed using a moving average filter over 1000 episodes from a single deterministic run; statistical variability is not shown.*

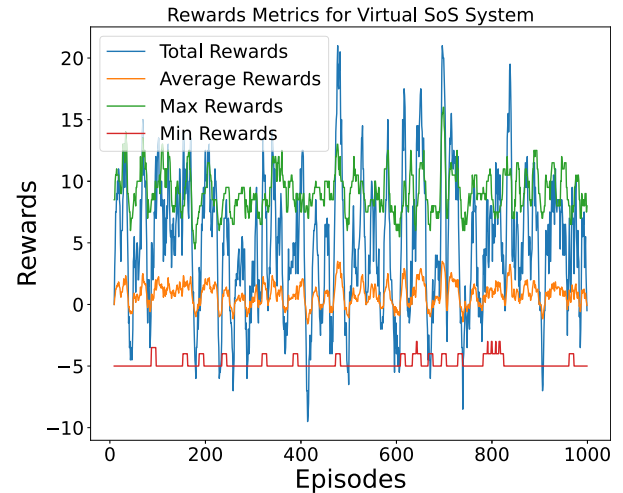


FIGURE 8. Reward metrics for the Virtual SoS system. Moderate fluctuations in total and maximum rewards indicate steady yet constrained performance. Persistent negative minimum rewards highlight limitations in agent adaptability under asynchronous, data-driven coordination. *Results are smoothed over 1000 episodes using a moving average filter from a single deterministic simulation; statistical error bars are not shown.*

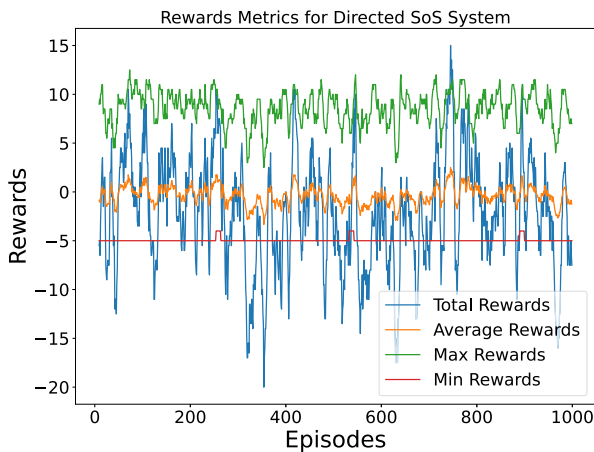


FIGURE 7. Reward metrics for the Directed SoS system. All reward curves show high volatility, with the minimum reward remaining persistently negative. These trends reflect challenges in centralized coordination and the agents' limited adaptability under a static control hierarchy. *A moving average filter was applied over 1000 episodes from a single deterministic run to smooth episodic fluctuations and highlight long-term reward dynamics; no error bars are included.*

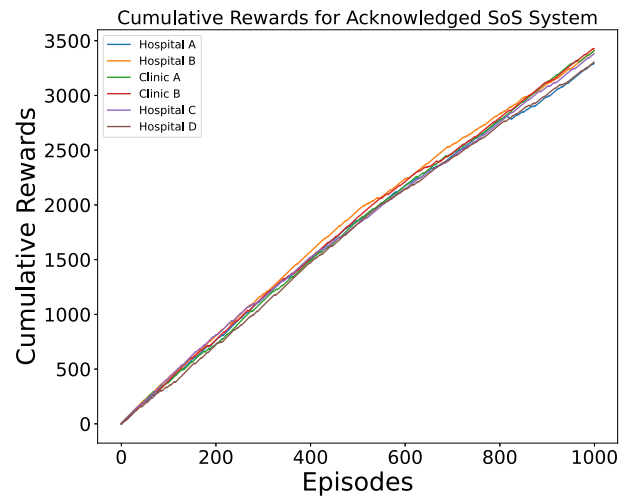


FIGURE 9. Cumulative rewards per agent in the Acknowledged SoS system. All agents demonstrate monotonic reward accumulation, converging near 3500 by episode 1000, indicative of robust semi-centralized coordination and consistent policy effectiveness across diverse agent roles. *Note: Results are generated from a single deterministic simulation and smoothed using a moving average filter over 1000 episodes; error bars are not shown.*

improvement, as depicted in Figure 9. The agents follow near-parallel, upward-sloping curves that converge around 3500 by episode 1000. This pattern reflects effective policy learning and steady reward accumulation across both physical systems (such as hospitals and clinics) and digital platforms. The uniformity in these trends further underscores the stabilizing effect of semi-centralized coordination.

Agents operating within the Collaborative SoS system demonstrate cumulative reward trajectories with varying slopes, as shown in Figure 10. By episode 1000, reward values span from 2000 to 2700, indicating that all agents experience learning gains over time. However, the differences

in curve steepness—particularly among telemedicine and wearable agents—point to uneven reward acquisition, likely influenced by asynchronous coordination mechanisms and heterogeneous local observations. Despite this variability, the overall upward trend suggests successful adaptation under decentralized control.

The cumulative reward trajectories under the Directed SoS system reveal notable disparities among agents, as illustrated in Figure 11. Unlike the consistent upward trends observed in the Acknowledged and Collaborative systems, agents in the

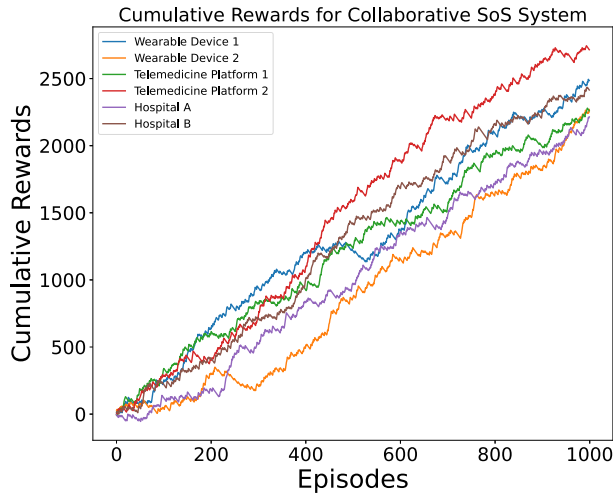


FIGURE 10. Cumulative rewards per agent for the Collaborative SoS. Agents include hospitals, telemedicine platforms, and wearable devices operating under a decentralized coordination scheme. The reward curves demonstrate heterogeneous but generally positive learning behavior.

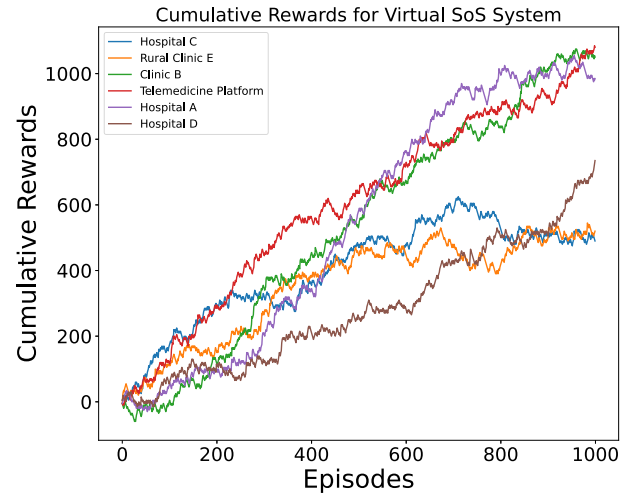


FIGURE 12. Cumulative rewards per agent for the Virtual SoS. Agents include hospitals, clinics, and telemedicine platforms operating asynchronously through data-driven coordination and remote monitoring.

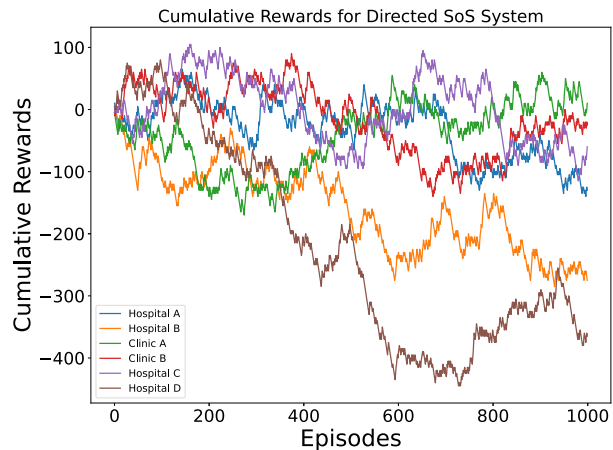


FIGURE 11. Cumulative rewards per agent for the Directed SoS. Agents represent hospitals and clinics operating under a centralized authority (see Table 1).

Directed configuration exhibit substantial divergence. While some hospitals accumulate modest rewards, others—such as Hospital C and Clinic A—experience sustained decline, with cumulative values dropping below -400 . These outcomes highlight the limitations of centralized control, where constrained adaptability and diminished local decision-making capabilities contribute to persistent underperformance. Since the Directed system lacks a learning mechanism, agents are unable to adjust their behavior in response to changing conditions—leading to accumulation of negative rewards over time.

The cumulative reward trends under the Virtual SoS configuration, shown in Figure 12, indicate gradual but steady policy improvement across all agents. While final values remain lower than those achieved in the Acknowledged and Collaborative systems, most agents converge between 700 and 1100 by episode 1000, reflecting stable yet

limited reward accumulation. Notably, the slower progress of HospitalC and ClinicB underscores the performance variability induced by asynchronous coordination and reduced control granularity in data-driven healthcare environments.

A comparative analysis of inter-agent cooperation, as shown in Figure 13, highlights distinct patterns across SoS configurations. The Acknowledged SoS maintains the highest and most stable cooperation levels, averaging nearly 5 per episode, reflecting strong alignment among semi-autonomous agents. Both the Collaborative and Directed systems exhibit moderate and fluctuating cooperation between 3 and 4, suggesting intermittent coordination challenges. In contrast, the Virtual SoS consistently lags, with cooperation counts remaining below 2, largely due to its reliance on loosely coupled, asynchronously operating digital agents. These trends underscore the importance of structural integration in fostering sustained multi-agent collaboration.

In summary, Acknowledged SoS consistently outperforms other systems in terms of Q-values, rewards, and cooperation, demonstrating superior adaptability and learning. Collaborative SoS performs well, particularly in wearable and telemedicine integration. Virtual SoS shows moderate but stable performance, while Directed SoS struggles with volatility and lower reward accumulation, highlighting challenges in central decision-making.

3) SYSTEM PERFORMANCE METRICS

This section analyzes the overall performance of each SoS system in optimizing resource allocation and agent decision-making based on cumulative rewards and Q-learning behavior.

As demonstrated in the comparison of cumulative rewards (Figure 9, Figure 10, Figure 11, and Figure 12), each system exhibited distinct learning dynamics throughout the simulation. The Acknowledged SoS system achieved the highest

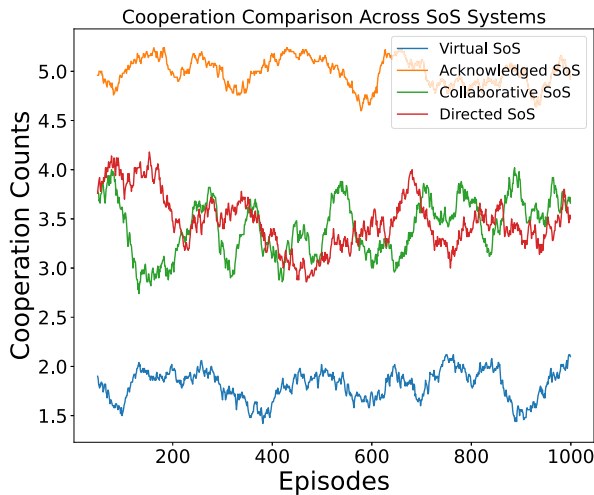


FIGURE 13. Cooperation counts comparison across SoS systems. The Acknowledged SoS consistently demonstrates the highest and most stable cooperation levels, followed by the Collaborative and Directed systems. The Virtual SoS maintains the lowest cooperation levels, attributed to its asynchronous coordination via loosely coupled remote agents. These trends highlight the relationship between system architecture and inter-agent coordination efficiency.

cumulative rewards, peaking around 3500 by the 1000th episode, indicating superior optimization of resources and agent coordination. The Collaborative SoS system followed closely, with cumulative rewards reaching approximately 2500. This demonstrates a relatively high efficiency in agent collaboration and decision-making, although with slightly more variability.

In contrast, the Virtual SoS system attained a more moderate cumulative reward of around 1000 by the end of the simulation, suggesting that the virtual coordination between telemedicine platforms and hospitals, while functional, was not as efficient as the physical hospital collaboration seen in other systems. The Directed SoS system, on the other hand, displayed significant challenges. Cumulative rewards for some agents, such as Hospital B and Clinic A, fell to negative values below -400 , highlighting inefficiencies in centralized decision-making.

The Q-learning trends support these findings. As seen in Figure 4, agents in the Directed SoS system consistently struggled to optimize decision-making, with Q-values fluctuating between -50 and 0 throughout the simulation. This indicates persistent challenges in adapting to resource allocation in a centralized control environment. In contrast, the Acknowledged SoS system exhibited strong and stable Q-value growth, reaching approximately 350 early in the simulation and maintaining this high level across episodes. The Collaborative SoS system also demonstrated positive Q-value trends, peaking around 250, though with more fluctuation than the Acknowledged system. Meanwhile, the Virtual SoS system showed more modest learning, with Q-values improving to around 100 by the end of the simulation.

These results indicate that decentralized systems, particularly the Acknowledged and Collaborative SoS, enabled more efficient learning and adaptation to resource allocation challenges. In contrast, the Directed SoS, constrained by centralized control, struggled to achieve the same level of efficiency and stability.

4) EXPLOITATION, EXPLORATION, AND LEARNING BEHAVIOR ACROSS SoS SYSTEMS

In multi-agent reinforcement learning, the balance between exploiting known strategies and exploring new ones is crucial for optimal system performance. This balance allows agents to refine their decision-making by either leveraging previously learned policies (*exploitation*) or searching for new, potentially better strategies (*exploration*). The results of this balance are evident in the behavior of agents across different Systems-of-Systems (SoS) configurations.

As shown in Figure 4, the **Acknowledged SoS** system consistently outperformed others in terms of Q-value evolution, with agents achieving the highest average Q-values, peaking at approximately 350 by episode 300. This performance is directly tied to the system's high *exploitation counts*, as illustrated in Figure 14, where agents maintained exploitation rates above 5.3 throughout the simulation. This indicates that agents in the Acknowledged SoS were highly efficient at leveraging learned policies to optimize resource allocation and decision-making, minimizing the need for exploratory actions.

Figure 14 compares exploitation counts across the four SoS configurations. The Acknowledged and Virtual SoS systems consistently maintain higher exploitation levels, averaging above 5.4 actions per episode, indicating stable policy convergence and reduced reliance on exploration. By contrast, the Collaborative SoS system exhibits lower exploitation rates, averaging around 4.6, with noticeable variance over time. This trend reflects agents' ongoing need to explore new strategies in response to decentralized, dynamically evolving environments. The Directed SoS system follows a similar pattern of reduced exploitation, suggesting that rigid centralization may limit agents' ability to consistently adhere to optimized policies.

Exploration behavior across all SoS configurations is shown in Figure 15. The Collaborative SoS exhibits the highest and most persistent exploration rates, averaging approximately 1.6 exploratory actions per episode. This trend reflects agents' ongoing search for improved policies within a decentralized and dynamic environment. In contrast, both the Acknowledged and Virtual systems display more stable and lower exploration levels, averaging below 1.0, indicative of early policy convergence. The Directed SoS shows moderate but erratic exploration behavior, marked by notable oscillations, which signal difficulties in stabilizing learning under rigid centralized control.

Among all configurations, the *Directed SoS* system encountered the greatest difficulty in balancing exploration and exploitation. As shown in Figures 14 and 15, agents

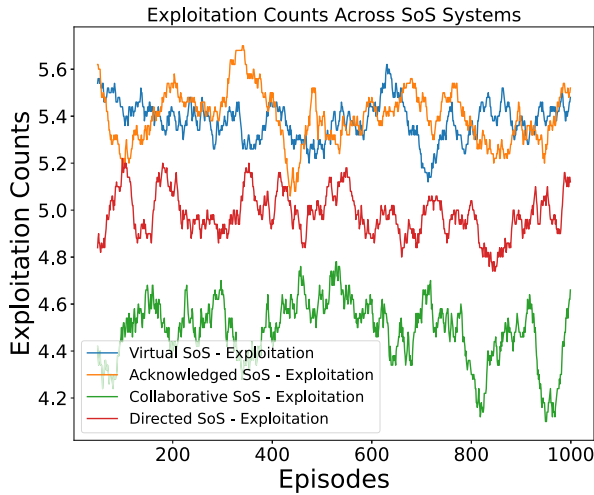


FIGURE 14. Exploitation counts across SoS systems. The Acknowledged and Virtual SoS configurations exhibit consistently higher exploitation frequencies—averaging above 5.4 per episode—indicating effective convergence to stable policies. In contrast, the Collaborative and Directed systems demonstrate lower and more variable exploitation counts, reflecting ongoing adaptation and less consistent policy adherence due to decentralized or rigid coordination structures.

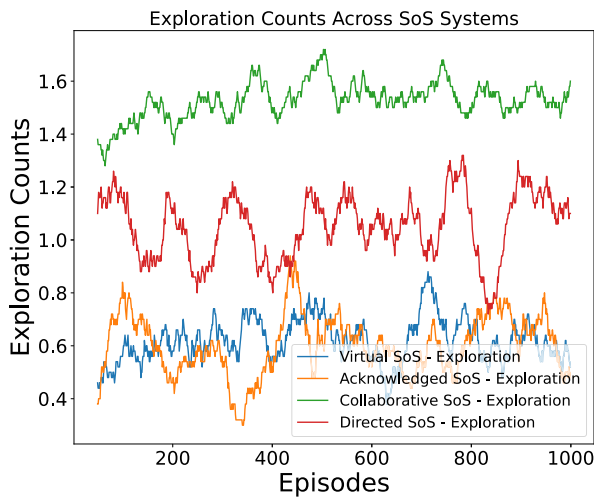


FIGURE 15. Exploration counts across SoS systems. Collaborative SoS agents maintain the highest exploration levels, indicative of ongoing adaptation under decentralized coordination. Lower exploration in Acknowledged and Virtual systems suggests early policy convergence, while Directed SoS shows intermittent exploration consistent with unstable learning.

operating under this model exhibited reduced exploitation rates and irregular exploratory behavior. This imbalance hindered effective learning, as agents struggled to consistently apply or refine policies. The rigid centralized control structure limited both adaptability and strategic discovery, ultimately contributing to diminished performance in terms of Q-value progression and cumulative reward accumulation.

The previously discussed cooperation counts (Figure 4) further reinforce these trends. The **Acknowledged SoS** displayed the highest agent coordination, averaging around 5.5, while the **Directed SoS** system showed lower cooperation levels. This reflects the benefits of decentralized systems in

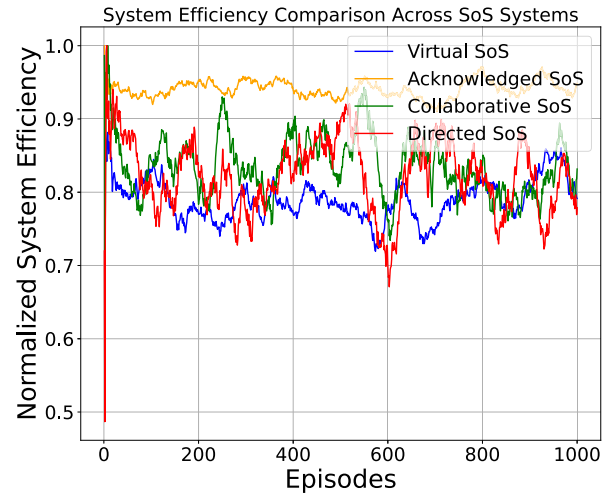


FIGURE 16. Smoothed comparison of normalized system efficiency across SoS systems. Acknowledged SoS exhibits the highest and most stable efficiency, highlighting the benefits of semi-centralized learning. Collaborative and Virtual systems perform moderately, with periodic fluctuations. Directed SoS demonstrates delayed convergence and volatility due to limitations of centralized control.

promoting inter-agent collaboration and learning, as opposed to the limitations imposed by a centralized decision-making structure in the Directed SoS.

5) SYSTEM EFFICIENCY COMPARISON ACROSS SoS SYSTEMS

Figure 16 presents the normalized system efficiency trends across all SoS configurations over 1000 episodes. The Acknowledged SoS maintains consistently high efficiency, averaging above 0.95, demonstrating effective semi-centralized coordination and rapid adaptation to dynamic resource demands. The Collaborative SoS exhibits moderate performance, fluctuating between 0.80 and 0.90, reflecting occasional coordination lapses under distributed control. The Virtual SoS achieves stable efficiency around 0.85, enabled by simplified coordination through digital interfaces but limited by the absence of physical resource reallocation. The Directed SoS shows persistent volatility and the lowest average efficiency, starting below 0.5 and stabilizing only after prolonged training, indicating inefficiencies associated with centralized decision-making under dynamic workloads.

The Acknowledged SoS consistently outperforms others, sustaining an efficiency level near 0.97. This indicates that agents operating under decentralized but coordinated policies adapt more effectively to evolving system demands.

The Collaborative SoS shows moderate variability, with efficiency oscillating between 0.80 and 0.90. While agents coordinate well across heterogeneous platforms, including wearables and telehealth, sporadic inefficiencies arise due to distributed learning.

The Virtual SoS remains relatively stable around 0.85, benefiting from the simplicity of remote interaction models. However, its lack of physical coordination among member systems constrains its overall optimization potential.

TABLE 3. Learned performance metrics of member systems under each constituent system.

Constituent System	Q-val	Util.	Coop.	Exploit.	Explore.
Directed	-36.92	381.37	4.00	5.08	0.88
Acknowledged	337.07	564.98	4.90	5.38	0.62
Collaborative	210.00	295.72	3.66	4.66	1.60
Virtual	80.17	30.08	2.10	5.48	0.52

In contrast, the Directed SoS begins with poor efficiency (below 0.5), improving only after extended training. The delayed convergence reflects challenges in centralized policy learning and slower adaptation to dynamic healthcare loads. To synthesize these findings, Table 3 reports final metrics aggregated across the last 10 episodes, including average Q-values, resource utilization, agent cooperation, and exploration-exploitation behaviors. These metrics reflect the learned behavior of constituent member systems operating under each SoS architecture.

Overall, the findings reaffirm the strategic benefits of decentralized coordination. Member systems operating under the Acknowledged and Collaborative constituent systems exhibit superior learning outcomes and resource synergy. In contrast, Directed constituent systems demonstrate delayed adaptation, highlighting the importance of local autonomy and distributed intelligence in MARL-enabled healthcare System-of-Systems architectures.

VI. DISCUSSION

This section discusses the implications of the results, compares our findings with prior work, identifies areas for future improvements, and addresses the key challenges in current research highlighted in earlier sections.

A. INTERPRETATION OF THE RESULTS

The performance of agents, as indicated by Q-value trends and cumulative rewards, demonstrates the effectiveness of decentralized decision-making frameworks in optimizing resource allocation across the SoS systems. Notably, the Acknowledged SoS system consistently achieved superior outcomes, with agents demonstrating a high level of cooperation (Figure 13), Q-values reaching 350 (Figure 4), and cumulative rewards nearing 3500 (Figure 9). These results indicate the robustness of a decentralized yet coordinated system, where agents independently manage resources while adhering to shared goals, leading to optimized performance.

In contrast, the Directed SoS system faced significant challenges, as reflected in its fluctuating Q-values and negative cumulative rewards for certain agents (Figure 11). Centralized control in a dynamic healthcare environment resulted in suboptimal resource allocation, limiting the system's ability to adapt to real-time changes in patient demand and resource availability.

B. IMPLICATIONS OF FINDINGS

The simulation results underscore the suitability of decentralized Systems-of-Systems (SoS) architectures—particularly

the Acknowledged and Collaborative models—for AIO/MT-enabled healthcare environments, where adaptability, autonomy, and efficient resource management are essential. In these systems, agents demonstrated the ability to learn and coordinate effectively, as reflected in high cooperation counts and stable exploitation behaviors (Figure 14). These behavioral trends suggest that decentralized agents can formulate locally optimal policies while still aligning with broader system objectives, contributing to improved system-wide outcomes. Such behavior is consistent with the principles of distributed decision-making frameworks previously advocated in healthcare system studies [4], [5].

Moreover, the performance of the Virtual SoS system, which maintained steady cumulative rewards despite minimal resource utilization (Figure 3, Figure 12), illustrates the potential of telemedicine and remote monitoring platforms to support sustainable healthcare delivery. These results imply that Virtual SoS configurations can be particularly advantageous in under-resourced or geographically isolated regions, where reliance on physical infrastructure is limited. The ability to sustain performance in such scenarios suggests a promising role for Virtual SoS models in enhancing healthcare access and resilience in constrained settings.

C. ADDRESSING KEY CHALLENGES IN CURRENT RESEARCH

1) SYSTEMS-OF-SYSTEMS (SoS) INTEGRATION

A key contribution of this research lies in addressing the long-standing challenge of integrating heterogeneous healthcare systems into a cohesive operational framework. The proposed MARL-based architecture facilitates coordination among distributed entities—such as hospitals, wearable devices, and telemedicine platforms—without requiring a monolithic control mechanism. Simulation results illustrate that Acknowledged and Collaborative SoS systems, both of which preserve local agent autonomy while fostering inter-agent coordination, significantly enhance resource sharing, diagnostic accuracy, and overall system performance. These benefits are particularly evident in the high cumulative rewards and cooperation counts observed in Figures 9 and 13. Compared to prior studies that focused on isolated system optimization [2], [3], this work demonstrates the operational advantages of integrated, learning-based coordination strategies.

2) HYPERPARAMETER TUNING AND REWARD STRUCTURE DESIGN

The performance of the Directed SoS system and its counterparts was sensitive to the tuning of key hyperparameters—including the learning rate α , discount factor γ , and exploration rate ϵ —each of which governs the agent's ability to adapt and converge within a dynamic environment. In the Directed SoS case, a fixed exploration rate of $\epsilon = 0.1$ encouraged agents to occasionally explore novel strategies, such as adjusting admission thresholds or

reallocating personnel. This was complemented by a 90% exploitation policy to retain learned behavior. As illustrated in Figure 15, Collaborative SoS maintained a higher exploration rate, averaging around 1.6 exploratory actions per episode, whereas Acknowledged and Directed systems exhibited lower exploration, favoring exploitative stability.

The reward function was also crafted to reflect key healthcare objectives. Agents received higher rewards for actions that aligned with operational goals—for instance, admitting patients with linked wearable telemetry or rebalancing limited staff. The Q-value evolution in Figure 4 confirms this alignment, where the Acknowledged SoS outperformed others with smooth convergence toward high-value policies. Meanwhile, Directed SoS agents struggled, accumulating negative or unstable cumulative rewards as shown in Figure 11.

3) SCALABILITY AND AGENT ADAPTATION

The MARL framework was designed to be inherently scalable across a diverse set of agent types and SoS configurations. Simulations confirmed that decentralized architectures, such as Acknowledged and Collaborative SoS, scaled effectively, allowing multiple agents to independently learn optimal policies for resource management under heterogeneous constraints. While the Directed SoS system did allow for centralized policy refinement, its agents exhibited limited adaptability and higher performance volatility, as evidenced by both fluctuating Q-values and lower cumulative rewards (Figure 4).

In contrast, decentralized agents in Acknowledged and Collaborative systems consistently demonstrated more efficient adaptation to stochastic environmental changes, enabling them to maintain higher reward trajectories. These results affirm the framework's robustness and scalability, particularly in environments characterized by complex, distributed decision-making and real-time demands.

D. COMPARISONS WITH PRIOR WORK

Our results align with existing literature that advocates for decentralized, collaborative systems in healthcare, particularly in managing complex resource allocation. Prior research on SoS integration focused on enhancing interoperability without fully leveraging MARL frameworks [2], [3]. By incorporating MARL into SoS architectures, this study offers a novel approach to optimizing resource use and patient outcomes, a gap previously underexplored [11].

Our findings also contrast with centralized models, like the Directed SoS, which showed inefficiencies, particularly in dynamic environments [5]. This suggests future research should enhance decentralized decision-making to handle healthcare uncertainties better.

E. DEPLOYMENT CONSIDERATIONS

While the simulation framework demonstrates the potential of MARL-based SoS coordination, several challenges must be addressed for real-world deployment. Privacy regulations,

such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR), require robust data handling and agent communication protocols to ensure patient confidentiality. The computational burden of MARL, especially in large-scale or real-time healthcare systems, necessitates efficient, scalable learning architectures. Furthermore, the reliance on inter-agent communication in decentralized systems may be impacted by network latency or limited infrastructure in certain settings. Addressing these challenges will require future integration with secure, federated learning systems and the development of bandwidth-efficient agent protocols to support robust, privacy-preserving coordination in operational environments.

F. LIMITATIONS AND POTENTIAL IMPROVEMENTS

Although the proposed MARL framework demonstrates strong performance, particularly in decentralized Systems-of-Systems (SoS) configurations, several limitations should be acknowledged. The use of synthetically generated data—while essential for maintaining ethical standards and modeling diverse scenarios—may not fully capture the complexity, noise, and unexpected correlations found in real-world healthcare operations. Future studies should validate the framework using empirical datasets, ideally drawn from hospital networks, telemedicine logs, or wearable device telemetry, to assess generalizability and robustness under actual clinical conditions.

One modeling limitation arises from agent variability across constituent systems. Although each simulation includes six agents, their identities and functional roles are adapted to reflect the coordination logic specific to the Directed, Acknowledged, Collaborative, and Virtual configurations. This design choice enhances architectural fidelity but limits direct cross-agent comparisons across constituent systems. Future extensions may consider normalizing agent roles or introducing abstraction layers to support more generalized and comparative evaluations across heterogeneous coordination models.

Furthermore, while decentralized agents proved effective in optimizing resource allocation, they inherently rely on communication to coordinate decisions. In large-scale or bandwidth-constrained settings, the communication overhead can introduce latency, affecting real-time responsiveness. This issue becomes particularly relevant for edge environments where infrastructure is limited. To mitigate this, future research could explore lightweight communication strategies or incorporate federated reinforcement learning to reduce inter-agent data exchange without compromising learning efficacy.

Lastly, the performance limitations observed in the Directed SoS system underscore the need for more advanced centralized control models that can dynamically adapt to changing system states. Centralized policies, when static or slow to update, are prone to bottlenecks and inefficiencies. Enhancing these systems with predictive modeling—such as

AI-driven demand forecasting or context-aware scheduling—could significantly improve responsiveness and reduce resource misallocation. Integrating hybrid architectures that blend centralized guidance with local autonomy may also offer a promising avenue for future exploration.

VII. CONCLUSION

This study presented a novel Systems-of-Systems (SoS) architecture integrated with a Multi-Agent Reinforcement Learning (MARL) framework to address the complex challenges of coordination, adaptability, and resource allocation in AIoMT-enabled healthcare environments. By modeling and evaluating four distinct SoS configurations—Directed, Acknowledged, Collaborative, and Virtual—the framework enabled a comparative analysis of how system-level structure influences learning performance, agent cooperation, and system-wide efficiency.

Simulation results demonstrated that decentralized coordination strategies—particularly within the Acknowledged and Collaborative constituent systems—consistently outperformed their centralized counterpart. The Acknowledged system achieved the highest cumulative rewards (approaching 3500) and sustained superior Q-value growth, exploitation efficiency, and inter-agent cooperation over 1000 episodes. In contrast, the Directed system, governed by centralized control policies, exhibited learning instability and suboptimal performance. Several agents in this configuration accumulated negative cumulative rewards, reflecting slower adaptation and the limitations imposed by rigid, top-down decision-making structures.

Several key conclusions emerge from this research:

- Decentralized architectures such as Acknowledged and Collaborative SoS exhibit enhanced adaptability and coordination, enabling agents to dynamically respond to real-time variations in healthcare demand and resource availability.
- Centralized systems, typified by the Directed SoS, are less responsive in dynamic environments and more susceptible to decision bottlenecks, leading to reduced cumulative rewards and cooperation counts.
- The success of the MARL framework depends heavily on the careful tuning of hyperparameters—especially the exploration-exploitation balance—which directly impacts agent learning efficiency and reward accumulation.
- The proposed framework demonstrates strong scalability, effectively supporting heterogeneous agent interactions among hospitals, telemedicine platforms, and wearable devices over prolonged simulation horizons.

While the results validate the effectiveness of decentralized MARL-based SoS in optimizing AIoMT healthcare coordination, the study is not without limitations. The use of synthetic data, though ethically essential, may not fully capture real-world complexity or edge-case scenarios. Future work will incorporate empirical datasets to assess the practical transferability of the learned policies. Moreover,

the reliance on frequent inter-agent communication in decentralized systems raises concerns in low-bandwidth or infrastructure-limited settings. Addressing these challenges through communication-efficient learning protocols and hybrid control architectures remains an open research direction. In future work, we aim to incorporate deep reinforcement learning techniques such as DQN or actor-critic models to better accommodate larger state spaces and partial observability in real-world healthcare environments.

Future research will extend this work in several concrete directions. First, we aim to evaluate the MARL framework using real-world hospital and telemedicine data to validate policy effectiveness under operational constraints. Second, to address scalability and partial observability in complex environments, we plan to incorporate deep reinforcement learning approaches, such as Deep Q-Networks (DQN) and actor-critic methods, and to explore continuous action spaces using algorithms like Deep Deterministic Policy Gradient (DDPG). Finally, we will investigate privacy-preserving and federated learning strategies to enable distributed, secure coordination in large-scale healthcare Systems-of-Systems.

In summary, this work underscores the transformative potential of MARL in enabling intelligent, scalable, and adaptable healthcare systems through SoS coordination. By integrating agent autonomy with shared system objectives, the framework lays the foundation for future advancements in AI-driven healthcare optimization, especially in settings characterized by distributed decision-making and resource uncertainty.

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